

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Winter Examination – 2022**

**Course: B. Tech.      Branch :Computer Science & Engineering      Semester :III**

**Subject Code & Name: (BTCOC304)Computer Architecture & Organization**

**Max Marks: 60**

**Date:**

**Duration: 3 Hr.**

***Instructions to the Students:***

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                    | (Level/CO) | Marks     |
|------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                        |            | <b>12</b> |
| A) Explain the different components of Central Processing Unit?                    | Understand | <b>6</b>  |
| B) Explain different types of memories with examples?                              | Remember   | <b>6</b>  |
| C) Explain with neat diagram computer components top- level view?                  | Analysis   | <b>6</b>  |
| <b>Q.2 Solve Any Two of the following.</b>                                         |            | <b>12</b> |
| A) Explain different types of instructions sets.                                   | Remember   | <b>6</b>  |
| B) Explain different addressing modes.                                             | Analysis   | <b>6</b>  |
| C) Explain the architecture of RISC and CISC processor.                            | Understand | <b>6</b>  |
| <b>Q. 3 Solve Any One of the following.</b>                                        |            | <b>12</b> |
| A) Explain the function of ALU.                                                    | Understand | <b>6</b>  |
| B) Briefly explain the following representations: sign magnitude, twos complement. | Synthesis  | <b>6</b>  |
| C) What are the four essential elements of a number in floating-point notation?    | Analysis   | <b>6</b>  |
| <b>Q.4 Solve Any Two of the following.</b>                                         |            | <b>12</b> |
| A) Explain types of semiconductor memories technologies.                           | Understand | <b>6</b>  |
| B) Explain memory hierarchy in computer system.                                    | Remember   | <b>6</b>  |
| C) Explain the working of optical memory.                                          | Understand | <b>6</b>  |
| <b>Q. 5 Solve Any One of the following.</b>                                        |            | <b>12</b> |
| A) Explain Programmed I/O module and Interrupt driven I/O.                         | Understand | <b>6</b>  |
| B) Explain Input/output organization of computer system.                           | Remember   | <b>6</b>  |
| C) Explain the Flynn's classification.                                             | Analysis   | <b>6</b>  |